

Separable Differential Equations

Def. A first order differential equation, $\frac{dy}{dx} = H(x, y)$, is called **separable** if we can write $H(x, y) = g(x)h(y)$.

In that case:

$$\frac{dy}{dx} = g(x)h(y)$$

$$\frac{1}{h(y)} dy = g(x) dx$$

then,

$$\int \frac{1}{h(y)} dy = \int g(x) dx .$$

Ex. Solve the initial value problem

$$\frac{dy}{dx} = -8xy; \quad y(0) = 4.$$

Notice that $H(x, y) = (-8x)(y) = g(x)h(y)$.

$$\frac{dy}{y} = -8x dx$$

$$\int \frac{dy}{y} = \int -8x dx$$

$$\ln|y| + c_1 = -4x^2 + c_2$$

$$\ln|y| = -4x^2 + c_3$$

$$e^{\ln|y|} = e^{-4x^2 + c_3}$$

$$|y| = e^{-4x^2} \cdot e^{c_3} = Ae^{-4x^2}; \quad (\text{general solution}).$$

$$y(0) = 4 \text{ so,}$$

$$|y(0)| = 4 = Ae^{-4(0)^2}$$

$$4 = Ae^0 = A$$

$$\text{So, } |y| = 4e^{-4x^2}$$

$$\text{But } y(0) = 4 > 0 \text{ so } y > 0 \text{ near } x = 0.$$

$$\text{Thus } y = |y| \text{ and}$$

$$y = 4e^{-4x^2} \quad (\text{particular solution}).$$

Ex. Solve the initial value problem

$$\frac{dy}{dx} = \frac{5-4x}{y(4y^2+2)}; \quad y(1) = 2.$$

$$\frac{dy}{dx} = \frac{5-4x}{4y^3+2y} = (5-4x) \left(\frac{1}{4y^3+2y} \right) = g(x)h(y).$$

$$(4y^3 + 2y)dy = (5 - 4x)dx$$

$$\int (4y^3 + 2y) dy = \int (5 - 4x) dx$$

$$y^4 + y^2 + c_1 = 5x - 2x^2 + c_2$$

$$y^4 + y^2 = 5x - 2x^2 + c_3 \quad (\text{general solution}).$$

One can't easily solve this equation for y in terms of x , so we leave the solution in this form. This equation represents a set of curves in the x - y plane

where $\frac{dy}{dx} = \frac{5-4x}{4y^3+2y}$ at every point (x, y) that fits the equation

$$y^4 + y^2 = 5x - 2x^2 + c_3.$$

Notice that if you differentiate this equation implicitly you will get

$$\frac{dy}{dx} = \frac{5-4x}{4y^3+2y}.$$

Now for the particular solution to the initial value problem we plug in

$$y(1) = 2 \text{ into } y^4 + y^2 = 5x - 2x^2 + c_3:$$

$$2^4 + 2^2 = 5(1) - 2(1)^2 + c_3$$

$$20 = 5 - 2 + c_3$$

$$17 = c_3$$

So the solution to the initial value problem is:

$$y^4 + y^2 = 5x - 2x^2 + 17.$$

Ex. Find the general solution to $y \frac{dy}{dx} - (8x^2 y)^{\frac{1}{3}} = 0$.

$$y \frac{dy}{dx} = (8x^2 y)^{\frac{1}{3}} = 2x^{\frac{2}{3}} y^{\frac{1}{3}}$$

$$\frac{y}{y^{\frac{1}{3}}} \frac{dy}{dx} = 2x^{\frac{2}{3}}$$

$$y^{\frac{2}{3}} dy = 2x^{\frac{2}{3}} dx$$

$$\int y^{\frac{2}{3}} dy = \int 2x^{\frac{2}{3}} dx$$

$$\frac{3}{5}y^{\frac{5}{3}} + c_1 = \frac{6}{5}x^{\frac{5}{3}} + c_2$$

$$\frac{3}{5}y^{\frac{5}{3}} = \frac{6}{5}x^{\frac{5}{3}} + c_3$$

$$y^{\frac{5}{3}} = 2x^{\frac{5}{3}} + \frac{5}{3}c_3$$

$$y^{\frac{5}{3}} = 2x^{\frac{5}{3}} + c_4$$

$$y = (2x^{\frac{5}{3}} + c_4)^{\frac{3}{5}}.$$

Ex. Find the particular solution to the initial value problem:

$$e^y \frac{dy}{dx} = 3e^{(3x-2y)}; \quad y(0) = \frac{1}{3} \ln(5).$$

$$e^y \frac{dy}{dx} = 3e^{(3x-2y)} = \frac{3e^{3x}}{e^{2y}}$$

$$e^{3y} \frac{dy}{dx} = 3e^{3x}$$

$$e^{3y} dy = 3e^{3x} dx$$

$$\int e^{3y} dy = \int 3e^{3x} dx$$

$$\frac{1}{3}e^{3y} + c_1 = e^{3x} + c_2$$

$$\frac{1}{3}e^{3y} = e^{3x} + c_3$$

$$e^{3y} = 3e^{3x} + c_4$$

$$3y = \ln(3e^{3x} + c_4)$$

$$y = \frac{1}{3} \ln(3e^{3x} + c_4) \quad (\text{general solution}).$$

$$y(0) = \frac{1}{3} \ln 5 \text{ so,}$$

$$\frac{1}{3} \ln 5 = \frac{1}{3} \ln(3e^0 + c_4) = \frac{1}{3} \ln(3 + c_4)$$

$$\ln 5 = \ln(3 + c_4)$$

$$5 = 3 + c_4$$

$$c_4 = 2$$

$$y = \frac{1}{3} \ln(3e^{3x} + 2) \quad (\text{particular solution}).$$

Population Growth and Continuously Compounded Interest

Both population growth and continuously compounded interest can be modeled based on the rate of change, $(\frac{dP}{dt})$, being a constant multiple of the amount $(kP(t))$. So:

$$\frac{dP}{dt} = kP; \quad \text{where } k \text{ is the annual growth rate.}$$

Separating variables we get:

$$\frac{1}{P} dP = k dt$$

$$\int \frac{1}{P} dP = \int k dt$$

$$\ln P + c_1 = kt + c_2 \quad (\text{since } P(t) > 0)$$

$$\ln P = kt + c_3$$

$$e^{\ln P} = e^{(kt+c_3)} = e^{kt} \cdot e^{c_3}$$

$$P(t) = c_4 e^{kt} \quad (\text{general solution}).$$

If $P_0 = P(0)$, then

$$P_0 = P(0) = c_4 e^0 = c_4$$

So, $P(t) = P_0 e^{kt} \quad (\text{particular solution}).$

Ex. The population of a town in 2010 was 100,000. The town's population in 2013 was 134,986.

- Find the annual growth rate.
- How long does it take for the population to double?

a) $P(t) = 100,000 e^{kt}$

$$P(3) = 134,986$$

$$134,986 = 100,000 e^{3k}$$

$$1.34986 = e^{3k}$$

$$\ln(1.34986) = 3k \quad \text{now using a calculator we get:}$$

$$.3 \approx 3k$$

$$k = .1 \quad \text{So the annual growth rate is 10\%}.$$

$$\text{b) } 200,000 = 100,000e^{.1t}$$

$$2 = e^{.1t}$$

$$\ln 2 = .1t$$

$$10 \ln 2 = t$$

$t \approx 6.93$ years for the population to double.

Radioactive decay also has the property that the rate of decay is proportional to the amount present so $\frac{dN}{dt} = -kN$, $k = \text{annual decay rate} \geq 0$, and $N(t) = \text{amount present} > 0$.

$$\frac{dN}{N} = -kdt$$

$$\int \frac{dN}{N} = \int -kdt$$

$$\ln N = -kt + c_1$$

$$e^{\ln N} = e^{-kt+c_1} = e^{-kt} \cdot e^{c_1}$$

$$N(t) = c_2 e^{-kt} \quad (\text{general solution})$$

If $N_0 = N(0)$, then

$$N_0 = N(0) = c_2 e^{-k(0)} = c_2.$$

So, $N(t) = N_0 e^{-kt}$ (particular solution).